MILITARY SPECIFICATION

GENERATOR SET, STEAM TURBINE, 1500-KW,

450-VOLT, 60-CYCLE, A.C.

1. SCOPE

.1 This specification covers steam turbine
ator sets for use on non-nuclear surface ships.

. APPLICABLE DOCUMENTS

.1 The following specifications, standards,
rawings of the issue in effect on date of invi-
for bids, form a part of this specification to
tent specified herein.

PECIFICATIONS

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.

MIL-H-194 - Heaters, Space, Electric: Blower and Turrett; and Heating Elements, Electrical, Strip-Naval Shipboard.

MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).

MIL-D-963 - Drawings, Production, Procedure for Procurement of.

MIL-V-1137 - Varnish, Electrical-Insulating (for Electro-Motive Equipment).

MIL-R-2729 - Regulator Sets, Voltage A.C. Generator, Naval Shipboard Use.

MIL-G-3087 - Generator Sets, Steam Turbine (Direct and Alternating Current) (Naval Shipboard Use).

MIL-G-3124 - Generator, Alternating Current, 60-cycle (Naval Shipboard Use).

MIL-M-9868 - Microfilming of Engineering Documents, 35mm, Requirements for.

MIL-M-15071 - Manual, Technical, for Mechanical and Electrical Equipment (Less Electronics).

MIL-P-15137 - Provisioning and Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).

MIL-C-15430 - Condensers, Steam, Surface, Naval Shipboard.

MIL-E-15465 - Ejector Assemblies, Air.

MIL-C-15726 - Copper-Nickel Alloy Rods, and Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate).

MIL-C-15730 - Coolers, Fluid, Industrial, Naval Shipboard: Lubricating Oil, Hydraulic Oil, and Fresh Water.

MIL-T-16049 - Tachometers: Electrical; Self-Generating; Mechanical, Fixed Mounting and Hand Held; and Vibrating Reed.

MIL-P-16298 - Preservation, Packaging, Packing, and Marking of Electric Machines Having Rotating Parts and Associated Repair Parts.

MIL-G-16983 - Gage, Pressure, Dial Indicating.

MIL-M-17059 - Motors, Alternating-Current, Franctional HP (Shipboard Use).

MIL-P-17286 - Preservation, Packaging, Packing, and Marking of Propulsion and Auxiliary Steam Turbines and Gears, (Includes Associated Repair Parts).

MIL-T-17523 - Turbine, Steam, General Auxiliary (Naval Shipboard Use).

MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

MIL-G-17859 - Gear Assembly, Speed Decreaser; and Gear Assembly, Reverse (Naval Shipboard Use).

MIL-D-18051 - Drawings Applicable to Vessels of the United States Navy; Microfilming of.

MIL-G-18997 - Gages, Pressure, Dial Indicating, Bourdon Tube.

MIL-P-19131 - Pumps, Rotary, Power Driven, Miscellaneous

MIL-C-19836 - Coolers, Fluid, Industrial, Air, Motor and Generator, Naval Shipboard.

MIL-B-21230 - Bronze, Nickel Aluminum: Castings, Ship Propeller Application.

FSC 6115

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STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.
MIL-STD-804 - Formats and Coding of Tabulating and Aperture Cards for Engineering Data Micro-Reproduction System.

DRAWINGS

BUREAU OF SHIPS

9000-S6201-74453 - Switchboard, Appliances, Switchboard Unit, Navy Standard.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, N.Y.)

AMERICAN SOCIETY FOR TESTING MATERIALS

A105 - Forged or Rolled Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service.
A285 - Low and Intermediate Tensile Strength-Steel Plates of Flange and Firebox Qualities (Plates 2 Inches and Under in Thickness.)
B143 - Tin Bronze and Leaded Tin-Bronze Sand Castings.
B148 - Aluminum Bronze Sand Castings.

(Application for copies should be addressed to American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania).

3. REQUIREMENTS

3.1 Complete turbine generator sets.-

3.1.1 Turbine generators.- The turbine generator sets shall be in accordance with Specification MIL-G-3087, except as otherwise specified herein.

3.1.1.1 The turbine generator sets shall have the following rating and characteristics:

Type required	Type II (Specification MIL-G-3087)
Steam pressure (at throttle)	As specified (see 6.1)
Steam temperature (at throttle)	As specified (see 6.1)
Exhaust pressure	As specified (see 6.1)
Parallel operation requirement	Parallel operation required.
Rating:	
Kilowatt	1500 (maximum continuous)
Voltage	450 Volts a.c.
Insulation	Class B or H
Type field	Salient pole
Phases	3
Generator speed	1200 r.p.m. or as otherwise approved to give lowest combined unit weight.
Enclosure	Water-air-cooled.
Type of Voltage regulator	Type II or III (Specification MIL-R-2729).
Regulator per switchboard	One main.
Repair parts	As specified herein.
Frequency	60 c.p.s.
Ambient temperature	50°C.

3.1.1.2 Each turbine generator set shall be of the package type comprised of the turbine, generator, reduction gear (if used), subbase, condenser, air ejector, motor driven lube oil pump and oil cooler mounted as an integral unit. Integral hand oil pumps for starting shall be included in the package. Components in the package shall be arranged to occupy a minimum of space and to reduce weight to the lowest practicable figure consistent with specification requirements. The package shall employ the principle of three-point support wherein the subbase is supported by the ship at three points only.

3.1.1.3 The turbine shall be suitable for continuous operation with steam inlet conditions and temperature swings as specified (see 6.1).

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3.1.1.4 The overall dimensions and weight of the complete turbine generator set including attached accessories shall not exceed the following:

Length - 14 feet, 10 inches
Width - 6 feet, 6 inches
Height - 13 feet, 6 inches
Weight - 45,000 pounds.

3.1.1.5 The combined unit shall be designed to withstand class HI shock in accordance with the values shown on figure 1.

3.1.1.6 The combined unit shall be designed to operate in accordance with this specification, to maintain satisfactory lubrication, and to avoid loss of oil, under the following conditions:

(a) When the combined unit is permanently inclined from the normal horizontal position as much as 5 degrees on either end and 15 degrees to either side.
(b) When the combined unit is momentarily inclined from the normal horizontal position as much as 10 degrees on either end and 45 degrees to either side.

3.1.1.7 Each complete generator set shall be capable of withstanding without damage to any components a 3-phase fully asymmetrical short circuit at the switchboard (assuming 25 feet of interconnecting cable from generator terminals to switchboard). The contractor shall submit a summation of calculations on the anticipated short circuit torque at the generator air gap, the elastic characteristics of all rotors, the shaft and gear loading characteristics, and the short circuit torque multiplying factor. This data shall be accompanied with the manufacturer's statement that all factors have been considered in designing for this short circuit torque condition and that all components will withstand this loading without damage. Data shall be submitted with the installation drawing for review.

3.1.1.8 The use of commercial grade basic material equivalent to the minimum characteristics and performances of the applicable Military Specifications is encouraged. Source inspection of basic material shall be kept to a minimum consistent with quality control as required to assure that construction and performance requirements have been met.

3.2 Turbine and gears.- The turbine shall be in accordance with Specification MIL-T-17523 except as otherwise specified herein.

3.2.1 With the exception of rotors and blades specified in 3.2.11 and 3.2.12, the design of all parts shall be based upon allowable stresses for the material used. The stresses for parts subjected to temperatures of 650°F. or above shall be based on appropriate safety factors and shall be determined by yield strength, stress rupture (40,000 hours) or 0.1 percent creep (40,000 hours), whichever is less. For temperatures below 650°F., allowable stresses and appropriate safety factors shall be based solely on yield strength.

3.2.2 Emergency power.- Turbine shall be capable of delivering rated power at the corresponding speed of the generator, with satisfactory operation in all respects, including governing, under emergency conditions as specified (see 6.1).

3.2.3 Welding and allied processes.- Welding and allied processes shall be in accordance with Standard MIL-STD-278.

3.2.4 Heat insulation and lagging.- Means for holding insulation shall be attached to the casing by the turbine manufacturer. The turbine manufacturer shall furnish a drawing, including a Bill of Material, showing application of insulating materials.

3.2.5 Lubrication and cooling.- Lubrication and cooling shall be as specified in Specification MIL-T-17523 except as specified herein.

3.2.5.1 The design shall be based on the use of Navy symbol 2190 TEP oil with a viscosity of approximately 180-200 Staybolt seconds universal (S.S.U.) at 130°F.

3.2.5.2 Each turbine shall have its own oil sump with a capacity double the flow per minute of the oil delivered to the control and lubrication systems when operating at rated speed. The sump shall be so located that oil shall drain by gravity from the bearings. Each sump shall be fitted with oil level indicator, with low, normal, and high levels clearly marked thereon. Internal heating or cooling coils are not permitted. No paint will be allowed on surfaces contacting oil.

3.2.5.3 A hand oil pump shall be provided to supply oil to all bearings prior to starting the turbine. The design shall be such that it will not be necessary to run this pump when the turbine is being shut down. In addition to the hand pump and the attached unit oil pump, an a.c. motor driven oil pump shall be provided for use during the start-up. The motor pump and controller shall conform to Specification MIL-P-19131 with operating characteristics as determined by the system requirements. This pump shall cut in and cut out automatically at predetermined oil pressures through the use of a pressure switch having adjustable pressure settings.

3.2.6 Flanges.- All flanges subject to inlet pressure (such as steam inlet, steam chest drain or steam ring drain) shall be as specified (see 6.1).

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3.2.7 The requirement "steam connections, together with steam chests, necessary valves, piping, governors and safety devices shall be so arranged that they need not be disturbed in lifting of the turbine casings, unless otherwise approved by the bureau or agency concerned" is not applicable.

3.2.8 The requirement "steam chests shall be located on the lower halves of turbine casings, unless otherwise approved by the bureau or agency concerned" is not applicable.

3.2.9 Sufficient clearances shall be provided around bolt heads and nuts to permit the use of ordinary tools. All casing and steam-chest joints shall be made metal-to-metal, without the use of sheet packing materials; flange faces may be lightly coated with a thin coating of boiled linseed oil, boiled linseed oil and graphite, or some form of joint cement approved by the bureau or agency concerned, or suitable material such as unvulcanized rubber sheet packing (not over 1/32 inch thick, and laid in strips not over 1/2 inch wide) which will flow sufficiently to prevent any appreciable separation of the flange faces when the joint is finally set up, and which will withstand the temperature conditions and permit reasonable ease in breaking and remaking joints. Flange faces shall be accurately finished to parallel surfaces.

3.2.10 Shaft journals shall be uniform and smooth to at least a 16 micro-inch finish.

3.2.11 The rotor shall be so designed that the average stress in the most highly stressed cross-section of the bladed rotor, under design operating conditions, shall not exceed a stress value corresponding to a minimum calculated factor of safety of 2.0 based upon:

(a) The yield strength (0.02 percent offset) of the material.
(b) The stress rupture of the material.

3.2.12 The blades shall be so designed that the average stress in the most highly stressed cross-section of the blades, under design operating conditions, shall not exceed a stress value corresponding to a minimum calculated factor of safety of 2.0 based upon:

(a) The yield strength (0.02 percent offset) of the material.
(b) The stress rupture of the material.

3.2.13 When installed on the turbine shaft, the thrust bearing shall be on the end subjected to the greatest variation in temperature unless otherwise approved by the bureau or agency concerned.

3.2.14 A gland seal manifold shall be provided. This manifold shall receive steam from a source of 150 p.s.i.g. 400° F. steam via a reducing valve set to maintain a pressure of not more than 30 p.s.i.g. at the inlet connection to the gland seal manifold. A valve shall be provided at this connection so that a pressure of 2 p.s.i.g. may be maintained in the manifold. This manifold shall also have a connection to each gland, and a connection to a stage of the turbine with pressure less than atmosphere. Valves shall also be provided at this latter connection and at the connection to the low pressure end gland. Regulation of pressure in the manifold shall be manual.

3.2.15 A trip throttle valve shall be provided. The valve shall be a quick and positive closing valve, normally full opened, and arranged to close when actuated either by the functioning of the overspeed trip or the back pressure trip (if used) or by hand. When the trip throttle valve has functioned to trip the turbine, it shall be necessary to reset the actuating devices to their normal operating position and to turn the throttle valve handwheel to the closed position before steam may again be admitted to the turbine. Design of these devices shall be such that resetting may be readily accomplished with the turbine operating at 75 percent of the normal operating speed, but not until after the throttle valve has been fully closed. The automatic operation of the trip throttle valve and actuating devices shall be positive, reliable, and entirely independent of the degree of opening of the throttle valve. The trip throttle valve shall be normally held open by oil pressure from a pump attached to the unit. It shall be tripped by dumping oil pressure.

3.2.16 A speed indicating device conforming to type IC/EFC or IC/EFD of Specification MIL-T-16049 shall be provided with the indicating meter mounted on the turbine gageboard. The meter shall cover a range of speed from 0 r.p.m. to 11 percent above rated speed. Provision shall be made for the use of a portable hand tachometer to set overspeed trips.

3.2.17 Any special or unusual onboard repair parts which may in the judgment of the manufacturer or Government inspector, be necessary for the proper maintenance of the units shall not be furnished without approval of the bureau or agency concerned.

3.2.18 In addition to the repair parts required, the stock component parts (Bureau of Ships Special Material) shall consist of bladed rotors complete with thrust collar and emergency governor. Each rotor shall be packed and stock numbered individually. The quantity to be furnished shall be as specified in the contract or order (see 6.1).

3.2.19 Steam rate shall not exceed that specified (see 6.1) at rated load and design steam and exhaust conditions. The turbine steam rate at 60 percent of rated load shall be stated in bid proposals. This steam rate shall be based upon rated inlet steam conditions, the vacuum obtainable at 60 percent load with rated condenser tube water velocity and

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75°F. condenser circulating water injection temperature.

3.2.20 Steam pressure gages shall be 4-1/2 inches in diameter, and shall be furnished in accordance with Specification MIL-G-18997 or MIL-G-16983, as applicable.

3.2.21 The governing synchronizing motor shall be a single phase, readily reversible, commutator less a.c. motor in accordance with Specification MIL-M-17059.

3.2.22 The turbine shall be subjected to the test specified in Specification MIL-T-17523 except that for the "Overspeed test" the maximum acceptable reading (double amplitude) of the vibration indicator shall not exceed 0.001 inch as measured on the bearing cap or 0.002 inch as measured on the rotating shaft adjacent to the bearings. These limits shall not be exceeded at any speed up to 25 percent above designed maximum operating speed. For the "Performance test, assembled unit" the time necessary to replace oil cooler or tube bundle need not be estimated by the Government inspector.

3.3 Gears.- Reduction gears shall be in accordance with Specification MIL-G-17859 except as specified herein.

3.3.1 The gear load factor "K" shall be 175 maximum.

3.3.2 The surface finish on tooth meshing areas shall not be rougher than 32 microinches root mean square unless otherwise specified by the bureau or agency concerned. This surface finish shall apply for measurement in any direction. Measurement shall be made for at least 80 percent of the face width on two or more representative teeth.

3.3.3 Pinions may be connected to turbine shafts or quill shafts by solid or dental couplings.

3.3.4 Auxiliary turbine gear cases shall be fitted with vapor vents having a total vent area of at least 2 square inches. Vapor vents shall be of the dry condenser type.

3.3.5 Shore based repair parts and stock components for auxiliary turbine driven gears are not required. Onboard repair parts shall be supplied in accordance with Specification MIL-T-17523.

3.4 Lube oil coolers.- The turbine generator lubricating oil coolers shall be designed and constructed in accordance with class 2 of Specification MIL-C-15730.

3.4.1 The cooler shall form an integral part of the turbine generator unit.

3.4.2 Mercurial thermometers and pressure gages shall be installed for each cooler to indicate lubricating oil inlet and outlet temperature and pressures respectively. Thermometer wells and pressure taps shall be located in the piping adjacent to the cooler, rather than on the cooler.

3.5 Condensers.- The condensers shall be designed and constructed in accordance with type II, class B of Specification MIL-C-15430 except as otherwise specified herein.

3.5.1 Inner tube sheets shall be steel, firebox quality of ASTM Publication A285-57T or grade II of ASTM Publication A105-58T.

3.5.2 When steel flanges are furnished, the construction of the waterbox shall be such that the steel does not come in contact with salt water. Such construction shall consist of either a solid metal facing welded to the flange and of the same material as the shell of the waterbox or an inlay formed by deposited weld metal of the same material as the weld of the waterbox shell to the flange. Such inlay shall extend from the inside periphery of the flange to the inner edge of the bolting holes and shall be a minimum of 3/8 inch thick after the face of the flange has been finally machined.

3.5.3 Material.- For the parts enumerated, the materials used in the construction of type II, class B condensers (for surface ships) shall be as shown in table I.

Table I - Material for type II condensers.

Part	Material
Shells, tube support plates and baffle plates	Steel plate, firebox quality of ASTM A285
Waterboxes	Copper-nickel alloy, composition 90-10 of MIL-C-15726, with flanges of copper-nickel alloy, composition 90-10 of MIL-C-15726; or steel, firebox quality of ASTM A285; or valve bronze, alloy 2A of ASTM B143; or gun metal, alloy 1B of ASTM B143; or bronze, aluminum, alloy 9C of ASTM B148; or bronze, manganese-nickel-aluminum, alloy 2 of MIL-B-21230.

3.5.4 Thermometers shall be supplied by the contractor.

3.5.5 An 8-1/2-inch vacuum gage conforming to type II, composition I, of Specification MIL-G-18997 shall be provided by the contractor for each condenser. (These gages will be mounted by the shipbuilder.)

3.5.6 The turbine generator condenser shall form a part of the package unit with the turbine generator.

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3.5.7 The turbine generator condensers shall be double pass and shall be supplied with cooling water from independent circulating pumps, one for each condenser. Pumps will be supplied by the shipbuilder.

3.5.8 With rated flow of 75°F. cooling water, the condenser shall be capable of maintaining a vacuum at its steam inlet of not less than 25 inches Hg (referred to a 30-inch barometer) when it is condensing the rated quantity of exhaust steam from the turbine and receiving the connected drains. With rated flow of cooling water, the condenser shall also be capable of maintaining this vacuum when it is condensing steam from the turbine corresponding to 80 percent of rated load on a generator, receiving drains from connected machinery, and condensing additional steam and receiving additional drains as specified (see 6.1).

3.6 Air ejectors.- The turbine generator condenser air ejector assemblies shall be twin two stage units with inter and after condenser designed and constructed in accordance with type A of Specification MIL-E-15465 and as specified herein.

3.6.1 The air ejectors shall be steam jet type and shall be designed for normal operation with 135 p.s.i., 450°F. steam. The air ejector shall also operate satisfactorily with steam at 690°F. (maximum).

3.6.2 Air ejector assemblies for turbogenerator assemblies shall be mounted on the condensers which they serve.

3.7 Generator.- The generator shall conform to Specification MIL-G-3124 except as otherwise specified herein.

3.7.1 Electrical insulation.- Insulation shall be class B or H and shall conform to Specification MIL-E-917. In no case shall untreated fiber glass insulation be utilized as ground insulation. Treated fiber glass insulation shall be utilized as ground insulation only when more than one layer is applied. The use of glass cloth backed mica as ground insulated is acceptable where the glass cloth is utilized solely as a backing for the mica insulation. The number of impregnations of each type of generator shall conform to the minimum requirements of the manufacturer's detail insulation drawings as approved by the bureau or agency concerned for that specific type of generator. In any case, there shall be not less than three complete independent impregnations of any electrical winding. One impregnation shall be defined as one complete cycle of treating and curing (such as one immersion and one baking) by a recognized method. However, the brushing or spraying method shall not be considered as one of the required impregnations unless it is applied as the final coat of varnish. All insulated windings shall receive at least two of the required impregnations after the windings have been installed on the corresponding machine subassemblies. Except as otherwise specified herein, all insulated electrical windings shall receive at least two of the required impregnations before assembly on machine parts. Impregnation before assembly on machine parts will not be required for windings which are to be inserted in semienclosed slots or other windings where the method of installation is such as to make this preliminary impregnation impractical. In such cases, all required impregnations shall be made after the windings are installed on the machine parts. If class H insulation is used, insulations suitability tests in accordance with the procedures detailed in Specification MIL-E-917 shall be conducted and reference to the approval letter shall be shown on the master drawing.

3.7.2 Painting.-

3.7.2.1 Exterior.- Except for shafts and identification plates, exterior surfaces of the generator shall be painted in accordance with Specification MIL-E-917.

3.7.2.2 Interior.- Electrical insulation, surfaces in contact with lubricating oil or grease, commutators, collector rings, brushes, bearings, and bearing surfaces shall not be painted. Also, peripheries of rotors and any other rotating part of the machine from which centrifugal force may cause the paint to be thrown on the winding when the machine is operating at rated load and speed and at normal temperature, shall not be painted; insulation varnish conforming to Specification MIL-V-1137 may be applied to such parts. All other parts that are not corrosion-resisting (see Specification MIL-E-917 for listing of corrosion-resisting materials and treatment) shall be painted in accordance with Specification MIL-E-917 except that one coat of enamel of light color will be sufficient.

3.7.3 Short circuit.- Generators shall be capable of withstanding without injury to any part, a three-phase short circuit at the terminals for a period of 30 seconds with constant field excitation corresponding to the value of ceiling excitation provided when the associated voltage regulator is in operation. The generator shall withstand a single-phase short circuit between any two terminals for a period of 30 seconds with constant field excitation, corresponding to the value of ceiling excitation provided when the associated voltage regulator is in operation.

3.7.4 Generator capacity rating.- The kilowatt rating shall be the net power available for continuous use. If the generator has any auxiliary equipment, such as a static excitation system that receives power from the generator and is considered to function as an essential part of the basic generator set,

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the generator shall have capacity in excess to the net power rating to furnish this power. Such capacity is considered to be the gross kilowatt rating.

3.7.5 Overload for steam-turbine-driven generators.- At rated voltage and frequency and at a power factor of 0.50, each generator shall be capable of delivering 150 percent rated current for 1 minute. If kw. overload rating is specified (see 3.1.1.1), each generator shall also be capable of delivering, at rated voltage, frequency and power factor, 125 percent rated current for 2 hours and 150 percent rated current for 5 minutes.

3.7.6 Efficiency.- The generator efficiency shall be not less than 95.0 percent at rated load.

3.7.7 Wave shape.-

3.7.7.1 Deviation factor.- The deviation factor of the open-circuit terminal voltage shall not exceed 6 percent on any or all phases.

3.7.7.2 Harmonic content.- From no load to full load, the root mean square value of the sum of all of the harmonic components in the output voltage wave shall not exceed 6 percent of the root mean square value of the rated output voltage. The root mean square value of any single component shall not exceed 3 percent of the root mean square value of the rated output voltage. Harmonics of 0.2 percent or less of the root mean square value of the rated output voltage may be disregarded in the determination of the harmonic content of the output voltage wave.

3.7.8 Balance before and after assembly.- The generator rotor shall be balanced dynamically, prior to assembly with its own frame and bearing, to "zero" remaining unbalance on the machine used. The actual unbalance, based on machine calibration, shall be recorded on the test form. After assembly with its frame and bearings, the rotor shall be balanced as necessary to limit the vibration (double amplitude) to 0.001 inch as measured on the bearing cap or 0.002 inch as measured on the shaft adjacent to the bearing(s). These limits shall not be exceeded at any speed up to 125 percent rated speed.

3.7.9 Reactances.- The transient reactance and subtransient reactance shall be as specified in table II.

3.7.10 Heaters.- Electric heaters suitable for 115 volt a.c. single phase operation shall be provided to insure against condensation of moisture on the winding while the generator is idle. The heaters shall be of the wattage required and shall conform to either of the following:

(a) A strip heater conforming to type S of Specification MIL-H-194 except that tests shall be performed only in accordance with 4.10.1 specified herein.

(b) A standard commercial tubular type heater element having a sheath of corrosion-resistant steel, nickel-copper alloy or other equally corrosion-resistant alloy, and which satisfactorily passes tests specified in 4.10.2.

Table II - Generator constants.

Capacity	Permissible subtransient reactance (minimum)	Permissible transient reactance (maximum)	Permissible voltage dip (2 per unit impedance) (maximum)
Kilowatts 1,500	.132	.25	Percent 13

Note.-Reactance values are saturated values and are expressed in per unit quantities.

3.7.11 The generator air cooler shall conform to the requirements of class 2 of Specification MIL-C-19836.

3.7.12 The Dynamic balance test and Radio interference tests are not required.

3.8 Excitation and regulation system.-

3.8.1 The excitation and regulation system shall be in accordance with type II or III of Specification MIL-R-2729. If a type II is used, an exciter with a vertical shaft is not acceptable.

3.8.2 If type III is used, the power rectifiers (and reactors if feasible) shall be mounted in the cooling air stream of the generators, and these rectifiers (and reactors) shall be readily accessible for inspection, removal, and disassembly or replacement. Load current transformers and associated excitation components (not mounted in the generator enclosure) shall be mounted in a drip-proof enclosure arranged for bulkhead or deck mounting. Voltage regulator components, including signal current transformers, shall be designed and arranged for mounting in the associated switchboard (shipbuilder furnished) and shall fit into the space available for voltage regulator components for the standard switchboard shown on Drawing 9000-S6201-74453, type G-7. Silicon rectifiers may be used, transistors shall not be used.

3.9 Technical manuals.- Technical manuals shall be in accordance with type III of Specification MIL-M-15071 except that an introduction section shall be included. The following equipment shall be covered in the manual:

Steam turbine and associated accessories.
Reduction gears.
Lube-oil coolers.
Generator plus generator air cooler.

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Voltage regulator and exciter.
Manual control means.
Temperature indicating and alarm devices.
Governor motor.
Special devices on equipment.
Condenser.
Air ejector.
Voltage regulator.

3.9.1 Technical manuals shall contain reduced size prints of all drawings approved for electric equipment, and a list of all drawings approved by the bureau or agency concerned indicating the manufacturer's and the bureau or agency concerned drawing numbers in parallel columns. Manuals shall be approved by the bureau or agency concerned prior to final printing.

3.9.2 If offset printing is used in the publication of the manual, a complete set of negatives shall, after completion of the manuals, be furnished to the bureau or agency concerned and shall remain the property of the Government. Regardless of the method of printing used, one glossy print negative of each halftone illustration included in the manual shall be furnished the bureau or agency concerned.

3.9.3 Two copies of the finally approved technical manual shall be packed with each turbine generator set at the time of shipment.

3.10 Drawings.- All drawings shall be in accordance with Specification MIL-D-963. The drawings to be furnished and the information contained thereon shall be in accordance with the individual equipment specifications or as otherwise specified herein.

3.10.1 Turbine and gear drawings.- The following turbine and gear drawings shall be furnished:

(a) Cross-sectional assembly (see 3.10.1.1).
(b) Sub-assembly (see 3.10.1.2).
(c) Details (see 3.10.1.3).
(d) Clearance diagram (see 3.10.1.4).
(e) Governing system diagram.
(f) Oil system diagram.
(g) Lifting and handling arrangement.
(h) Lagging and insulation.
(i) Repair parts and tool list.
(j) List of drawings. The list of drawings shall contain a complete listing of all applicable drawings with the exception of drawings for standard commercial items.
(k) Gland seal and leakoff diagram.
(l) Oil piping assembly.
(m) List of material (see 3.10.1.5).

3.10.1.1 Cross-sectional assembly.- This drawing shall be of the layout type and shall include the following:

(a) Longitudinal sectional and other views so that sub-assemblies and major parts may be readily identified.
(b) Material of principal parts.
(c) Major dimensions not included on installation drawing.
(d) Water-removal features and casing drains.
(e) Table of stage pressures and temperatures at 1/4, 1/2, 3/4, and full load.
(f) Bearing data, including loads, surface speeds and pressure angle.
(g) Complete equipment performance curves.
(h) Critical speed.

3.10.1.2 Sub-assembly drawings.- The drawings shall be provided as necessary to show details of sub-assemblies not included in the cross-sectional assembly. Included in these drawings shall be one drawing or drawing list to describe the completely bladed and assembled turbine rotor with all rotating parts such as thrust collar and overspeed trip. This drawing shall state that the rotor is to be dynamically balanced and shall indicate the overspeed trip setting.

3.10.1.3 Detail drawings.- Details of standard commercial items such as nuts, bolts and anti-friction bearings shall not be furnished when they can be described by catalog number, size and so forth, on the cross-sectional assembly drawing or a sub-assembly drawing.

3.10.1.4 Clearance diagram.- The clearance diagram shall indicate the maximum and minimum acceptable clearances for all critical areas such as thrust and sleeve bearings, blading and packing.

3.10.1.5 List of material.- A list of all materials used, identified by drawing and piece number and manufacturer's part number, shall be furnished as a separate drawing.

3.10.2 Combined unit installation drawing.- Installation drawing(s) shall be in accordance with Specification MIL-G-3087 except as otherwise specified herein.

3.10.2.1 In addition to the drawings required by the individual equipment specifications, an installation drawing including the following shall be submitted:

(a) Dimensional outlined drawing of the complete generator set. The drawing shall include all connections, electrical and mechanical weights, and other data necessary to clearly describe the equipment. The drawing shall also include an outline of the identification plate showing all plate markings and the location of the plate.

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(b) Piping diagrams, showing recommended connections for oil, water, steam, and exhaust necessary for proper installation and operation of the generator set. The size, type and dimensions (for shipbuilding connections) of flanges including outside diameter, thickness, number of holes, bolt diameter, bolt circle and face shall be shown.
(c) Type number, ratings (volts, amperes, r.p.m., kw., etc.), and name of manufacturer, of all electric equipment.
(d) Outline and complete description of circuit breaker trip switch.
(e) Center of gravity of major components and center of gravity of the combined unit (wet and dry).
(f) Location, service and range of all gages and thermometers. The turbine gage board shall also be shown on the drawing.
(g) Any additional drawings required for the proper installation, operation, and maintenance of the equipment furnished.
(h) A complete performance table.
(i) A graphic presentation of allowable forces and moments on the steam inlet and exhaust connections.
(j) Estimated quantities of steam and air leakage at leakoff connections, dismantling dimensions, and contractual data.

3.10.3 Final drawings.- Final drawings shall consist of the following:

(a) One set of blueprints or black and white prints of the final approved electrical drawings.
(b) Microfilm in accordance with Specification MIL-D-18051 for all electrical and mechanical drawings.
(c) One set of card mounted microfilm for turbine and gear drawings. The microfilm shall be 35mm, filmed and processed in accordance with Specification MIL-M-9868, except as specified herein:
1. Selection - Card mounted microfilm, type II, class 1.
2. Size and weight of paper - In accordance with figure 1 of Standard MIL-STD-804.
3. Color - Buff.
4. Information - Cards shall contain the information shown on figure 2 of Standard MIL-STD-804.
5. Key punching - Required, but shall be approved by the bureau or agency concerned.
6. Aperture opening - Location on card shall be as shown on figure 1 of Standard MIL-STD-804.
7. Submittal - One complete set of card mounted film shall be forwarded as specified (see 6.1). These cards may be forwarded as a complete set (not later than roll submission) or in partial sets.

3.10.4 All preliminary drawings (unless approved master drawings are on file) shall be submitted to the bureau or agency concerned via the cognizant inspector for approval. The installation drawing shall be submitted for approval within 90 days from the date of contract. Drawings for the turbine, gear, lube-oil coolers, generator, exciter, and control equipment shall be submitted for approval within 120 days from the date of the contract.

3.11 Onboard repair parts.- Onboard repair parts shall be furnished in accordance with the individual equipment specification and as specified herein based on the number of equipments per ship specified in the contract or order.

3.11.1 Standard Navy stock numbers shall be obtained for all parts in accordance with Specification MIL-P-15137.

3.12 When any specification which forms a part of this specification requires that the product be subjected to and pass qualification tests, only products which have been approved for listing on the applicable Qualified Products List on the date of invitation for bids or date of initiation of purchase action shall be utilized. In the event no Qualified Products List has been issued, the contractor shall request instructions as to what testing will be required to determine whether the product meets the requirements of the specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Tests shall be conducted on the turbine generator units in accordance with Specification MIL-G-3087. Test results on the first generator and combined turbine-generator unit shall be forwarded to the bureau or agency concerned for approval prior to shipment of the units. At least 60

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days prior to testing, a test agenda shall be submitted to the bureau or agency concerned for approval.

4.3 All steam fittings and connections which may be subject to boiler pressure, including steam chests, throttle valves, and nozzle chambers, shall be tested hydrostatically after assembly to a pressure as specified (see 6.1).

4.4 A wave shape test at full load is not required.

4.5 Steam consumption shall be measured at 100 percent, 75 percent, 60 percent and 25 percent load with steam minimum rated pressure. Consumption shall be determined by measuring the condensate either by weighing or using water meters calibrated to the satisfaction of the inspector.

4.6 The generator set shall be operated continuously for 50 running hours at full rated kw. No adjustment shall be made on the set during the run. Data as required during the 1 hour continuous load test shall be recorded. The test shall be run with rated inlet steam pressure.

4.7 Periodic test data shall include the field current of the generator at ceiling excitation under single and three phase short circuit conditions with voltage regulator operative and generator field hot.

4.8 The ambient noise level for the audio noise test need not be guaranteed provided that the test is conducted when the ambient is at a minimum.

4.9 During the governor tests required by Specification MIL-G-3087, the no load voltage for sudden load change may be set higher than 105 percent, if necessary, to maintain rated voltage and load after load is applied. In addition, load valve lift data and load speed data may be taken in two separate runs.

4.10 Heater tests.-

4.10.1 Strip type.- The strip type heater shall be tested as specified in Specification MIL-H-194 except that the shock and vibration tests are not required. The tests shall be conducted at the place of manufacture.

4.10.2 Tubular type.- Each tubular type heating element shall be subjected to the following tests by the manufacturer:

(a) Wattage.- The wattage shall be measured and shall be within ±5 percent of the rated value.

(b) Water immersion.- The heating element and end seals shall be immersed in water for 10 minutes, after which the terminals shall be wiped dry and the dielectric and insulation resistance tests conducted.

(c) Insulation resistance.- An insulation resistance test shall be conducted using a potential of 500 volts d.c. The insulation resistance shall be not less than 100 megohms.

(d) Dielectric test.- The heating element shall be subjected to and shall withstand for a period of 1 minute a dielectric test voltage of twice the rated plus 1000 volts applied between a terminal and the sheath.

4.11 Inclined test.- The combined turbine generator set shall be operated at rated speed, with or without load, for at least 30 minutes in each of the following positions:

(a) Shaft inclined, forward end down 5 degrees.
(b) Shaft inclined, forward end up 5 degrees.
(c) Shaft horizontal, base inclined to right 15 degrees.
(d) Shaft horizontal, base inclined to left 15 degrees.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of shipboard repair parts.-

5.1.1 Generator set.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during handling and direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or order number, contractors name and destination.

5.1.2 Shipboard repair parts.- Shipboard repair parts shall be processed as follows:

5.1.2.1 Mechanical repair parts.- Mechanical repair parts shall be preserved and packaged level

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A, packed level C and marked levels A and C respectively, in accordance with the requirements of Specification MIL-P-17286.

5.1.2.2 Electrical repair parts.- Electrical repair parts shall be preserved and packaged level A, packed level C and marked levels A and C respectively in accordance with the requirements of Specification MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment requirements.-

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.1).

5.2.1.1 Preservation and packaging.-

5.2.1.1.1 Turbine generator set.-

5.2.1.1.1.1 Level A.-

5.2.1.1.1.1.1 Turbine, gears and couplings shall be preserved and packaged in accordance with the level A requirements of Specification MIL-P-17286.

5.2.1.1.1.1.2 Generators and other electrical components shall be preserved and packaged in accordance with the level A requirements of Specification MIL-P-16298 or MIL-E-17555 as applicable. When the size and mounting arrangement of the generator attached to the turbine (see 3.1.1.2) does not permit enclosure of the generator within a water-vaporproof barrier as required in a method II package, the alternate method of packaging specified in Specification MIL-P-16298 may be applied.

5.2.1.1.1.1.3 Condensers, air ejectors and coolers shall be preserved and packaged in accordance with the level A requirements of the applicable item specification.

5.2.1.1.1.1.4 Pumps shall be preserved and packaged in accordance with the domestic shipment and storage or overseas shipment requirements of specification MIL-P-16789.

5.2.1.1.1.2 Level C.- Preservation and packaging of turbine generator sets shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.2.1.1.2 Repair parts.- Repair parts shall be processed as follows:

5.2.1.1.2.1 Mechanical repair parts.- Mechanical repair parts shall be preserved and packaged in accordance with the level A requirements of Specification MIL-P-17286.

5.2.1.1.2.2 Electrical repair parts.- Electrical repair parts shall be preserved and packaged in accordance with the level A requirements of Specification MIL-E-17555.

5.2.1.2 Packing.-

5.2.1.2.1 Turbine generator sets.-

5.2.1.2.1.1 Assembled sets.-

5.2.1.2.1.1.1 Levels A and B.- When the package set (see 3.1.1.2) is shipped as a complete package, each set shall be anchored, blocked, braced and cushioned and enclosed in a sheathed crate conforming to the requirement of Specification MIL-C-104 and appendix thereto.

5.2.1.2.1.2 Disassembled sets.-

5.2.1.2.1.2.1 Levels A and B.- When the package set (see 3.1.1.2) is disassembled for shipment, the disassembled items and equipments shall be packed level A or B as specified (see 6.1) in accordance with the requirements of the applicable item or equipment specification.

5.2.1.2.2 Repair parts.-

5.2.1.2.2.1 Levels A and B.- Repair parts shall be packed level A or B as specified (see 6.1) in accordance with the applicable packaging specification (see 5.2.1.1.2). Unless otherwise specified (see 6.1) repair parts shall be packed separately and shipped concurrently with the basic equipment.

5.2.1.2.3 Unpacking instructions.- In addition to any special marking required by the contract or order, unpacking instructions of equipment, assembled or disassembled sets, shall be provided to prevent possible damage during removal. When practical, one set of these instructions shall be placed in a sealed waterproof envelop prominently marked "Unpacking Instructions" and securely affixed to the outside of the shipping container in a protected location. If the instructions cover a set of equipment packed in multiple containers, the instructions shall be affixed to the number one container of the set.

5.2.1.2.4 Technical manuals.- Technical manuals shall be packaged and packed for the level specified (see 6.1) in accordance with the manual specification (see 3.9).

5.2.1.2.5 Drawings.- Drawings shall be packaged and packed for the level specified (see 6.1) in accordance with the drawing specification (see 3.10).

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5.2.1.3 Marking.- Interior packages and exterior shipping containers shall be marked for shipment in accordance with the requirements of the applicable item specification and Standard MIL-STD-129.

5.2.1.3.1 Special marking.- Shipping containers shall be stencilled as follows:

CAUTION - THIS EQUIPMENT MAY BE DAMAGED UNLESS UNPACKING INSTRUCTIONS ARE CAREFULLY FOLLOWED. UNPACKING INSTRUCTIONS ARE LOCATED (state where located).

When practical, this marking shall be applied adjacent to the identification marking on the side of the container.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Steam pressure at throttle (see 3.1.1.1).
(b) Steam temperature at throttle (see 3.1.1.1)
(c) Exhaust pressure (see 3.1.1.1).
(d) Steam inlet conditions and maximum temperature swings for which the turbine must be suitable (see 3.1.1.3).
(e) Emergency conditions for turbine (see 3.2.2).
(f) Requirements for flanges subject to inlet pressure (see 3.2.6).
(g) Quantity of stock component parts (see 3.2.18).
(h) Steam rate (see 3.2.19).
(i) Amount and condition of additional steam and amount and temperature of additional drains which condenser must handle with generator at 80 percent rated load (see 3.5.8).
(j) Whether a gland exhaust ejector conforming to type C of Specification MIL-C-15465 is required, or whether after condenser surface shall be increased sufficiently to handle gland exhaust in conjunction with a shipbuilder furnished gland exhaust fan.
(k) Whether, although zinc anode fittings on the condenser are to be provided in accordance with specifications, the zinc anodes proper shall be omitted.
(l) That one set of card-mounted film shall be forwarded to the Bureau of Ships, Attn: Code 641 (see 3.10.3(d)).
(m) Hydrostatic test pressure for steam fittings and connections (see 4.3).
(n) Level of preservation, packaging and packing required (see section 5).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Navy - Bureau of Ships
(Project 6115-N052Sh)

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MIL-G-22665(SHIPS)

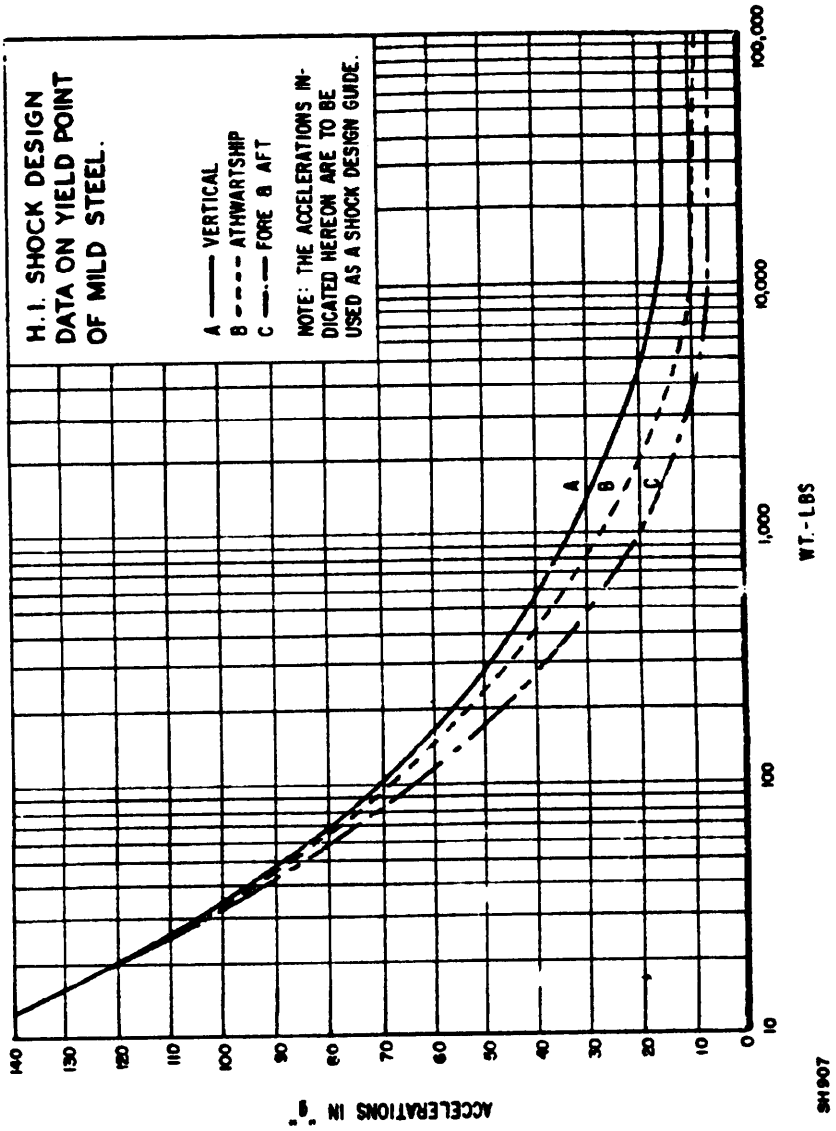


FIGURE I. - H.I. SHOCK DESIGN DATA

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